

HARSH RAJESH SHAH

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EDUCATION

University of Southern California

M.S. in Computer Science - Coursework in Algorithms, Robotics, Distributed Systems

Los Angeles, CA

Aug. 2025 – May. 2027

JSS Science and Technology University

B.E. in Computer Science

Mysuru, India

Aug. 2019 – Aug. 2023

PROFESSIONAL EXPERIENCE

Fidelity Investments

Bengaluru, India

Software Engineer

Aug. 2023 – June 2025

- Improved **system scalability and performance** by migrating legacy Java-based APIs to Python's FastAPI, while maintaining existing Spring Boot microservices, resulting in a **40%** reduction in response times
- Designed authentication flow using **Microsoft MSAL** with token caching and SSO, reducing login time by 30% while meeting enterprise security compliance requirements
- Directed the end-to-end migration from **Amazon EC2 to Amazon EKS**, containerizing microservices and implementing CI/CD pipelines to boost scalability by **3 times**, halve deployment time, and streamline continuous delivery operations
- Engineered a **proof-of-concept converter** to automatically translate **Informatica SQL mappings into SnapLogic pipelines**, automating **80%** of manual ETL migration tasks and validating a reusable framework for enterprise-wide adoption
- Developed a **Python-native ETL framework** to synchronize data across heterogeneous systems, eliminating third-party connectors and achieving fully automated, high-throughput data transfers with greater maintainability and fault tolerance

Fidelity Investments

Bengaluru, India

Software Engineering Intern

Feb. 2023 – June 2023

- Developed a centralized **Python API marketplace** for frequently used data requests, enabling users to **fetch data via validated access tokens** and extend **common base classes** for seamless integration into custom applications
- Designed and implemented a **frontend interface using Angular** to automate previously manual workflows, improving usability, reducing effort, and enabling faster, visual data-driven decision-making for business users

GE Healthcare

Bengaluru, India

Research Intern

Aug. 2022 – Nov. 2022

- Developed an application to automatically recognize actionable text in error logs and service manuals using **custom Named Entity Recognition (NER) pipelines** and reducing manual analysis time by over **60%**

PROJECTS

Soft-QD Reinforcement Learning Framework | MuJoCo, JAX, PPO

- Architected a **Soft Quality-Diversity (Soft-QD)** pipeline using PPO as the underlying policy optimizer to evolve a diverse archive of high-performing agents in MuJoCo Playground.
- Leveraged **JAX-based vectorization** to parallelize thousands of concurrent physics environments, significantly accelerating the collection of behavioral descriptors for the QD archive
- Designed a **custom reward shaping scheme** combining task reward with a diversity bonus, enabling the archive to cover a broad range of locomotion behaviors while preserving high policy performance

Parallel Redis Performance Prototype | C++, pthreads, mutex/lock-free structures, YCSB

- Increased system throughput and multi-core scalability** by re-engineering the single-threaded Redis core into a partitioned multi-threaded architecture to eliminate CPU bottlenecks.
- Optimized performance under skewed workloads by developing dependency-aware command queues to balance "hot key" distribution, measured via **YCSB across read/write/mixed workloads**

TECHNICAL SKILLS

Programming: Python, C, C++, Java, JavaScript, TypeScript, SQL

Frameworks/Tools: FastAPI, NodeJS, Angular, React, Spring Boot, Informatica, Jira, Agile/Scrum

ML/Robotics: JAX, Reinforcement Learning, PyTorch, MuJoCo/MJX, Brax

Cloud/Databases: AWS, Azure, PostgreSQL, MongoDB, Redis

DevOps: Git, Docker, Kubernetes, Jenkins